# A calculus of grounded measurement

how a DOME turns raw physical energy into the two signals the nervous system actually uses

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This figure is a **boundary object**: a neuroscientist, a biomechanist, a computer-vision engineer, and a control theorist all read the same pipeline but name its parts differently. The little algebra below gives them one precise, shared way to read it; nothing here assumes prior background. The one idea to carry through: **every box is a bundle of channels (degrees of freedom), each channel carries its own status, and every arrow is a typed operation.** Crucially, the only operation that can turn a well-measured input into a corrupted downstream feature is **projection** everything else either passes information through or combines it. That single fact is what lets the chart say not just **what** is missing, but **which downstream signal it breaks** and **which instrument would fix it.**

### 1 · The three ingredients

***Energies — what a sensor is physically sensitive to***

A sensor does not perceive objects; it converts one form of physical **energy** into numbers, and that choice fixes what it can and cannot know.

- $\mathcal{P}$ **photons** — light. Cameras, the eye tracker, and the terrain scanner live here.
- $\mathcal{I}$ **inertial** — the specific force and angular rate felt by a small body (an IMU's accelerometer + gyroscope).
- $\mathcal{M}$ **mechanical contact** — the forces and torques where the body meets the ground (a force plate).

***Values are channel vectors — not single grades***

A reading is a **typed value**, and a value is a **bundle of channels** (degrees of freedom, DOF). Status lives **per channel**, so no value is uniformly good or bad. The types:

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|---|---|
| $\mathbb{R}^n$ | $n$ numbers — e.g. a 3-D point, or a 6-number **wrench** (three forces + three torques). |
| reference frame | a coordinate system rigidly attached to a thing (eye, head, room) — distinct from a **video** frame. Position only means anything relative to one. |
| $\mathrm{SO}(3)$ | an **orientation** — three channels (e.g. for the eye: elevation, abduction, torsion). |
| $\mathrm{SE}(3)$ | a full **pose** — orientation **and** position, six channels; a rigid body's complete placement. |
| $q_{ab}$ | a **relative pose**, "$a$ in $b$" — e.g. eye-in-head is the eye's orientation relative to the skull. |
| $\mathfrak{G}$ | scene **geometry** — the surrounding surfaces and the groundplane. |

Worked all the way through, the **eye's** contribution to retinal input is five channels in two families — orientation {elevation, abduction, **torsion**} and optics {pupil, **accommodation**}. Three are measured well today; two (torsion, accommodation) are not. That per-channel split is the whole correction.

***Operators — how channels move***

Three operations combine values. Only one of them can move information **between** channels.

- $\oplus$ **compose poses** — **transparent.** Chain relative poses; the shared inner reference frame telescopes, $q_{ab} \oplus q_{bc} = q_{ac}$. It carries each input channel's status through **unchanged** and injects no new uncertainty — exact once the reference frames are jointly calibrated.
- $\bowtie$ **fuse** — **union of measured subspaces.** Combine two estimators of one quantity with complementary error (camera: accurate-imprecise $\bowtie$ IMU: precise-drifting). The fused value is measured on the **union** of what either measures — the point of complementary fusion.
- $\Pi$ **project** — **the only injector.** A forward model mapping a state onto an observable, via its Jacobian. A downstream channel is ● **measured** iff it loads **only** on measured DOF, and ◐ **inferred** iff it loads on **any** unmeasured DOF. This is where priors enter and where reachability is decided.

Two more relations sit at the ends of the chain. $\mathcal{E} \vdash s$ — **transduce** — is the one physical step: a sensor **yields** ($\vdash$) a raw signal $s$ (image, inertial stream, force) from an energy $\mathcal{E}$; everything after it is $\to$ — **derive** — data turned into data. And $v \approx v^\star$ says $v$ estimates the true latent $v^\star$. When a needed DOF is unmeasured, we compute the **conditional** estimate under an explicit **prior** on it; the estimate is trustworthy on the subspace the measured DOF span and biased on the complement by however far the true DOF strays from the prior — and that bias lands in **named, disjoint channels**, not smeared across the output.

***The two axes — what a channel mark means***

Replace the single grade with a status on two orthogonal axes:

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|---|---|
| **present** | ● **measured** directly and well ◐ **inferred** — reconstructed under an explicit prior; only as good as the prior holds. |
| **build** | ★ **build target** — an unmeasured DOF a committed instrument targets, because it is **load-bearing** ◇ **could build** — a feasible instrument, but nothing needs it yet. |

A channel can be both at once: torsion is ◐★ (inferred under a prior now; a build target), accommodation is ◐◇ (inferred now; buildable but not yet load-bearing). Nothing here is "unreachable at any price" — the marks track **instruments and priors**, never math difficulty.

## 2 · The three paths, per channel

*Path A — Vision: from photons to the retinal stimulus*

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|---|---|---|---|
| Eye tracker ⟶ eye keypoints | $\mathcal{P} \vdash$ image $\to$ keypoints | ● | transduce photons into an **image** ($\vdash$), then **derive** ($\to$) pupil/iris/limbus keypoint tracks. Mature. |
| Eye kinematics (5 channels) | elev ● abd ● torsion ◐★ pupil ● accom ◐◇ | | the **estimate** from those keypoints: eye-in-head. Orientation {elevation, abduction} and pupil are measured; **torsion** (roll about the line of sight) and **accommodation** (focal state) are unmeasured — inferred under priors. |
| Scene geometry | $\mathcal{P} \vdash$ depth $\to \mathfrak{G}$ | gnd ● · surf ◐ | groundplane is known/controllable (LED floor, actuated terrain — **manipulable**); distal surfaces are ◐. |
| Gaze in world | $g = q_{\text{eye}} \oplus q_{hw}$ | orient ● · tors ◐★ | $\oplus$ is transparent: gaze inherits the eye's channels unchanged (plus head pose from Path B). **This is why the eye and body cannot be separate instruments — gaze literally contains head pose.** |
| Retinal image & optic flow | $\rho = \Pi(g, \mathfrak{G}, \dot{x}_h) \approx \rho^\star$ | see below | $\Pi$ maps the input DOF onto the retinal channels; worked out in §2.5. **Afferent** input to the visual system — **Desideratum #1**. |

*Path B — Body: two motion-capture streams fused into kinematics*

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|---|---|---|---|
| Camera mocap ⟶ body keypoints | $\mathcal{P} \vdash$ images $\to p : \mathbb{R}^{3\times K}$ | ● | transduce photons into **images**, then **derive** $K$ body keypoint positions by triangulation — accurate in absolute placement, but **imprecise** (jitter). |
| IMU mocap | $\mathcal{I} \vdash$ accel, gyro $\to R : \mathrm{SO}(3)^S$ | rate ● · abs ◐ | body-worn IMUs: angular **rate** is measured precisely; absolute orientation is its drifting integral ⟶ ◐. |
| Body kinematics (Hybrid) | $X = p \bowtie R : \mathrm{SE}(3)^S$ | transl ● · orient ● | $\bowtie$ unions measured subspaces: camera positions anchor IMU drift, IMU rate smooths camera jitter ⟶ translation and orientation both ● once fused. The head-in-world $q_{hw}$ is just the head segment's entry of $X$. The **fusion instrument itself** is a ★ build target (synchronization), not the arithmetic. |

*Path C — Kinetics: from contact forces to the muscle drive*

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|---|---|---|---|
| Force plates & insoles | $\mathcal{M} \vdash$ force $\to w : \mathbb{R}^6$ | ● | transduce contact into a **force** reading, then resolve the ground reaction **wrench** $w$ (3 forces + 3 torques) and centre of pressure — near-direct. |
| Inverse dynamics | $\tau = \mathrm{ID}(X, w; \mathfrak{I})$ | ◐ | Newton–Euler mechanics from motion $X$, wrench $w$, and an **inertial model** $\mathfrak{I}$ (segment masses/inertias, BSP). Exact operator; the joint-torque channels are ◐ because $\mathfrak{I}$ is a prior, not a measurement. |
| Muscle forces & activation | $a = \mathrm{M}(\tau) \bowtie \mathrm{EMG}$ | model ◐ · EMG ● | the model estimate $\mathrm{M}(\tau)$ is ◐ (muscle **redundancy** ⟶ solved under an optimization prior); EMG senses activation **directly** (●). Two estimates of one latent ⟶ a fusion and a cross-check (§4). **Efferent** drive — **Desideratum #2**. |

***Cross-links — why the three paths are one instrument***

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|---|---|
| **head pose** (B⟶A) | gaze needs head-in-world from body kinematics. |
| **self-motion** (B⟶A) | retinal optic flow needs the head/eye translation $\dot{x}_h$, again from B. |
| **kinematics** (B⟶C) | inverse dynamics needs the same body motion $X$. |

***2.5 · Worked example — the retinal node is not one grade***

Pull the retinal observable back through $\Pi$. It has two channel families.

**Geometric (first-order optic flow).** The local flow field decomposes into four differential invariants: **divergence** (looming / time-to-contact), **curl** (rotation about the viewing axis), and two **deformation** (shear) components. Torsion is eye rotation **about the line of sight** — it is precisely the **curl** generator. So unmeasured torsion contaminates **exactly the curl channel**; divergence is invariant to it, and the deformation magnitude is invariant (only its axis rotates).

**Optical (photometric). Pupil** (measured) sets the aperture ⟶ retinal illuminance and depth-of-field: ●. **Accommodation** (unmeasured) sets the focal plane ⟶ the defocus-blur map: ◐◇.

So the retinal node is:

div ● def ● **curl** ◐★ (torsion) illum ● **defocus** ◐◇ (accommodation)

Two missing DOF, two **disjoint** corrupted features, two distinct future instruments — far more honest, and more useful, than one ⋆ on the box.

### 3 · The propagation law

There is no scalar "meet over the cone," and operators carry no grade. A node's channel-status is the **pushforward** of its inputs' channel-supports through its operator:

$\oplus$ carries supports through unchanged · $\bowtie$ unions the measured subspaces · $\Pi$ maps DOF $\longrightarrow$ channels by its Jacobian

Read it as: **pull an observable back through Π; whichever input DOF a channel loads on decides its mark —** ● if it touches only measured DOF, ◐ if it touches any unmeasured DOF (and it inherits that DOF's ★ / ◇). Because $\oplus$ and $\bowtie$ never inject uncertainty, **every** ◐ in the chart traces to a specific unmeasured DOF at a specific Π — which is exactly the thing an instrument can fix.

### 4 · The loop this unlocks — the real thesis

This is the payoff, and it is **selective**: it says **when** a missing DOF matters and when it can be ignored.

1. **Estimate** the downstream quantity under an **explicit prior** on the unmeasured DOF. (Retinal flow reconstructed with an eye tracker giving elevation + abduction only; torsion set by Listing's law.)
2. **Find which output channels drive the behaviour or science.** (Curl comes out as a load-bearing feature of the flow field.)
3. **Pull those channels back through Π: do they depend on an unmeasured DOF?** (Curl $\longleftarrow$ torsion. Yes.)
4. **Decide. Yes** $\longrightarrow$ the prior is load-bearing; that DOF is your next instrument (build the torsion-resolving tracker). **No** $\longrightarrow$ the missing DOF does not touch what matters here; you are already as good as possible and can ignore it honestly (pure looming / time-to-contact lives in divergence $\longrightarrow$ torsion irrelevant).

The subtle point: step 2's finding that **curl matters** is **simultaneously** a proof that **the torsion prior is load-bearing** — because curl's error **is** the gap between true torsion and Listing's law. The analysis does not just produce an estimate; it **justifies building the instrument.** (Matthis 2022 is the worked instance.)

### 5 · Endpoints, and a built-in observability check

The chart terminates at exactly two boxes because there are exactly two places where the measurable body meets the nervous system: the **retinal stimulus** (**afferent** — what **enters** the visual system) and the **muscle drive** (**efferent** — what **leaves** the motor system). Everything upstream is machinery for reaching those two with **known** error.

Muscle drive is special: it is both a **projection target** ($a = \mathrm{M}(\tau)$, inferred under the redundancy prior) **and directly sensed** (EMG). Two independent estimates of one latent give a consistency check on the Π from torque, and a candidate $\bowtie$ fusion — the same "a downstream projection target that is also directly measured" pattern that makes the eye story rigorous. Wherever the chart offers both a projected and a measured route to a channel, it can validate itself.

***Closing the loop — one instant in time***

Those two peripheral signals are the **input** to the **central nervous system** (cortical and subcortical, reciprocally coupled), whose **efferent** output is a motor command that re-aims the eye and re-drives the body. The whole chart is therefore a single instant $t$: the command is realised at $t + \delta t$, changing what every sensor measures on the next frame. The figure is one slice of a loop that repeats each $\delta t$ — preceded by the infinitesimal before it and followed by the one after. That recurrence is why the measurement problem is worth solving: each instant's estimate is the state on which the nervous system acts to produce the next.